

Project Overview

Title: Designing the Network Solution – SOHO LAN

Project 1 - Part 1: Basic Network Design

Content:

- ▶ Role: Network Engineer for a small startup (7 employees)
- ▶ Objectives: Design a simple LAN connecting essential devices
- ▶ Scope: Internet connectivity, IP addressing, and network topology

Network Requirements

Title: Hardware and Devices to be Connected

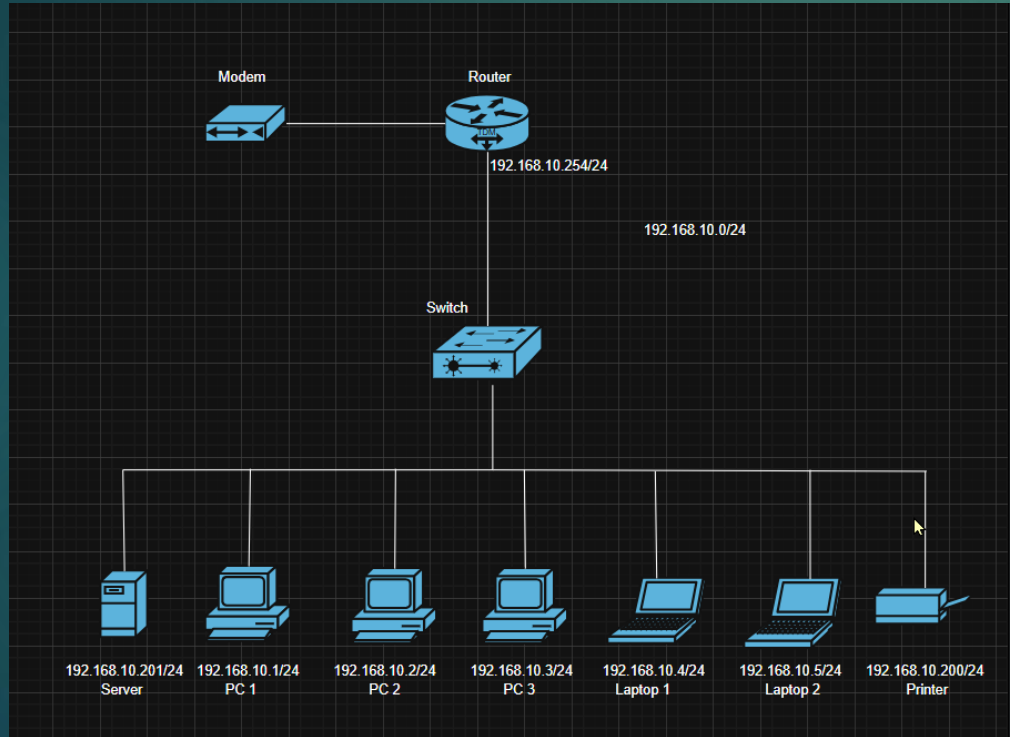
Content:

- ▶ 1x Router
- ▶ 1x 24-port Switch
- ▶ 1x Server
- ▶ 3x PCs
- ▶ 2x Laptops
- ▶ 1x Network Printer
- ▶ 1x Modem

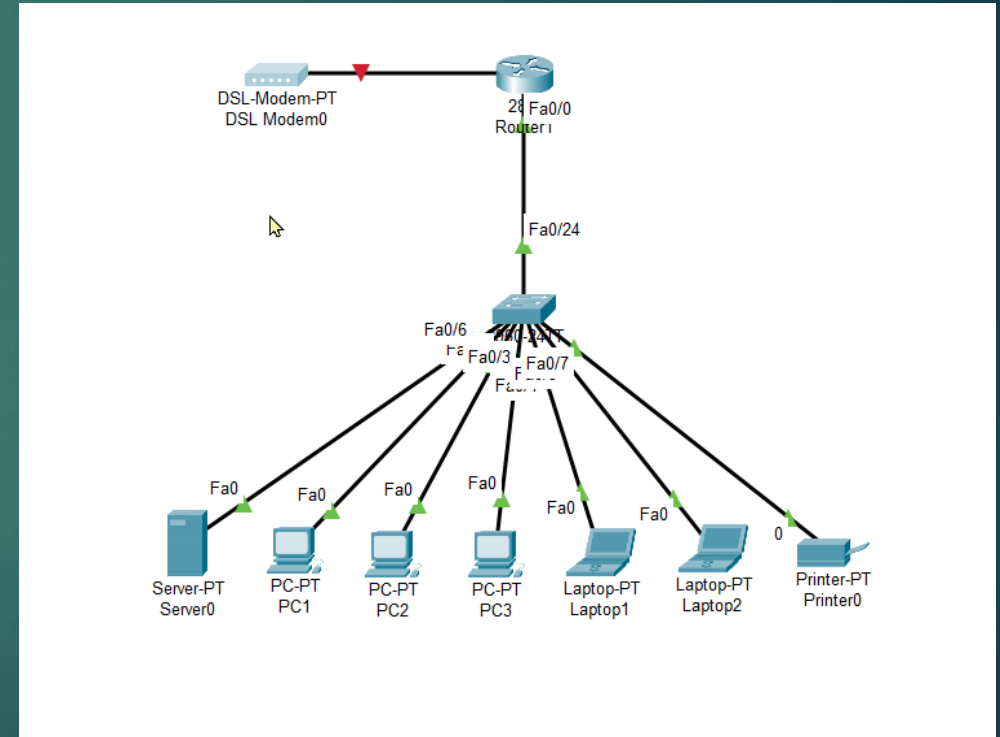
Project 1: Designing the Network Solution

Part 1: Basic Network Design

Simple LAN Topology Overview



DrawIO



Cisco Packet Tracer

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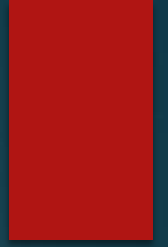
Part 1: Basic Network Design

IP Addressing Table

Devices	IP Address	Interface	Default Gateway
Router	192.168.10.254/24	FastEthernet0/0	NA
Switch	NA	NA	NA
Server	192.168.10.201/24	FastEthernet0	192.168.10.254
Network Printer	192.168.10.200/24	FastEthernet0	192.168.10.254
PC 1	192.168.10.1/24	FastEthernet0	192.168.10.254
PC 2	192.168.10.2/24	FastEthernet0	192.168.10.254
PC 3	192.168.10.3/24	FastEthernet0	192.168.10.254
Laptop 1	192.168.10.4/24	FastEthernet0	192.168.10.254
Laptop 2	192.168.10.5/24	FastEthernet0	192.168.10.254
Modem	NA	NA	NA

Project 1: Designing the Network Solution

Part 1: Basic Network Design



Internet Connectivity

- ▶ Research and choose a reliable ISP that could offer a appropriate bandwidth (Fibre-Optics).
- ▶ Subscribe to a plan that align with company usage and budget.
- ▶ Connect the modem to the router for all devices to access to the internet.
- ▶ Assigned static IP address for server and network printer to ensure always reachable and also easy to manage, in case of troubleshooting.
- ▶ Ensure router's firewall is up to block unwanted traffic to keep our internal network secure.

Part 2: Network Expansion Overview

Title: Network Expansion – HQ and Branch

Content:

- ▶ Business growth after 3 years, new branch added
- ▶ Network redesign: From 1 LAN to 3 LAN
- ▶ Tool: Cisco Packet Tracer

Project 1: Designing the Network Solution

Part 2: Network Expansion

Segmented LAN Design

Title: Segments Breakdown

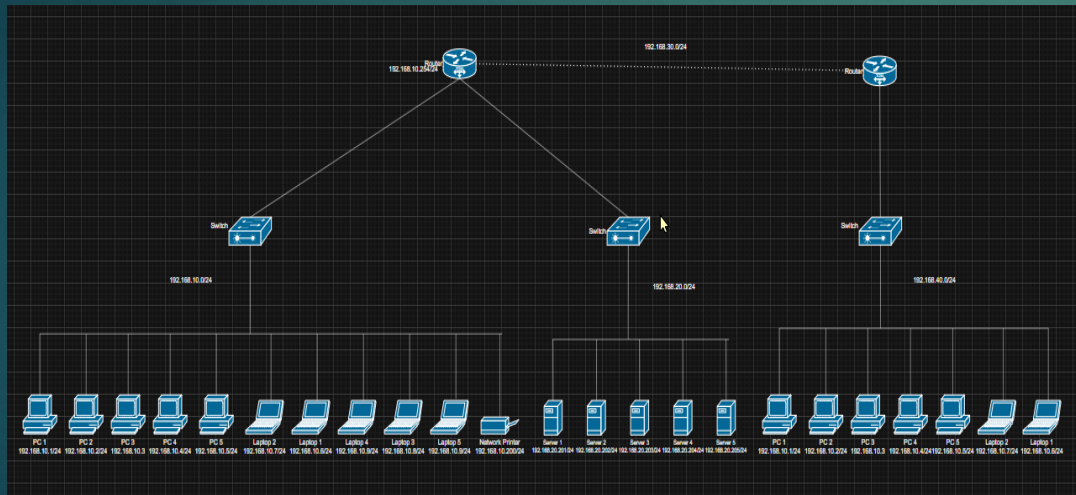
Content (Table):

Segment	Devices
HQ LAN 1	1 Router, 1 Switch, 5 PCs, 5 Laptops, 1 Printer
HQ LAN 2	1 Switch, 5 Servers
Branch LAN	1 Router, 1 Switch, 5 PCs, 2 Laptops

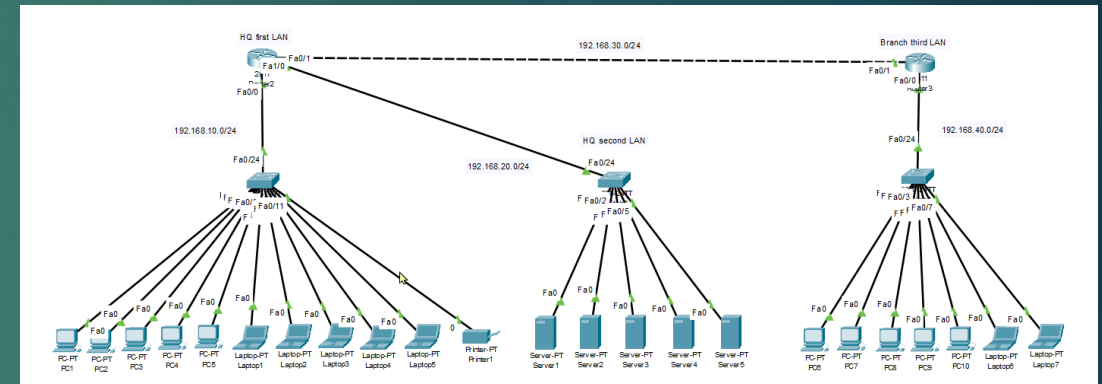
Project 1: Designing the Network Solution

Part 2: Network Expansion

Updated Network Diagram



DrawIO



Cisco Packet Tracer

Project 1: Designing the Network Solution

Part 2: Network Expansion

Extended IP Addressing Table

Devices	IP Address	Interface	Default Gateway
Router	192.168.10.254/24	FastEthernet0/0	NA
	192.168.20.254/24	FastEthernet1/0	NA
	192.168.30.254/24	FastEthernet0/1	NA
Switch	NA	NA	NA
PC 1	192.168.10.1/24	FastEthernet0	192.168.10.254
PC 2	192.168.10.2/24	FastEthernet0	192.168.10.254
PC 3	192.168.10.3/24	FastEthernet0	192.168.10.254
PC 4	192.168.10.4/24	FastEthernet0	192.168.10.254
PC 5	192.168.10.5/24	FastEthernet0	192.168.10.254
Laptop 1	192.168.10.6/24	FastEthernet0	192.168.10.254
Laptop 2	192.168.10.7/24	FastEthernet0	192.168.10.254
Laptop 3	192.168.10.8/24	FastEthernet0	192.168.10.254
Laptop 4	192.168.10.9/24	FastEthernet0	192.168.10.254
Laptop 5	192.168.10.10/24	FastEthernet0	192.168.10.254
Printer	192.168.10.200/24	FastEthernet0	192.168.10.254

Devices	IP Address	Interface	Default Gateway
Switch	NA	NA	NA
Server 1	192.168.20.201/24	FastEthernet0	192.168.20.254
Server 2	192.168.20.202/24	FastEthernet0	192.168.20.254
Server 3	192.168.20.203/24	FastEthernet0	192.168.20.254
Server 4	192.168.20.204/24	FastEthernet0	192.168.20.254
Server 5	192.168.20.205/24	FastEthernet0	192.168.20.254

HQ LAN 2

HQ LAN 1

Project 1: Designing the Network Solution

Part 2: Network Expansion

Extended IP Addressing Table

Branch

Devices	IP Address	Interface	Default Gateway
Router	192.168.40.254/24	FastEthernet0/0	NA
Switch	NA	NA	NA
PC 6	192.168.40.1/24	FastEthernet0	192.168.40.254
PC 7	192.168.40.2/24	FastEthernet0	192.168.40.254
PC 8	192.168.40.3/24	FastEthernet0	192.168.40.254
PC 9	192.168.40.4/24	FastEthernet0	192.168.40.254
PC 10	192.168.40.5/24	FastEthernet0	192.168.40.254
Laptop 6	192.168.40.6/24	FastEthernet0	192.168.40.254
Laptop 7	192.168.40.7/24	FastEthernet0	192.168.40.254

HQ to Branch Router

Devices	IP Address	Interface	Default Gateway
HQ Router	192.168.30.1/24	FastEthernet1/0	NA
Branch Router	192.168.30.2/24	FastEthernet0/1	NA

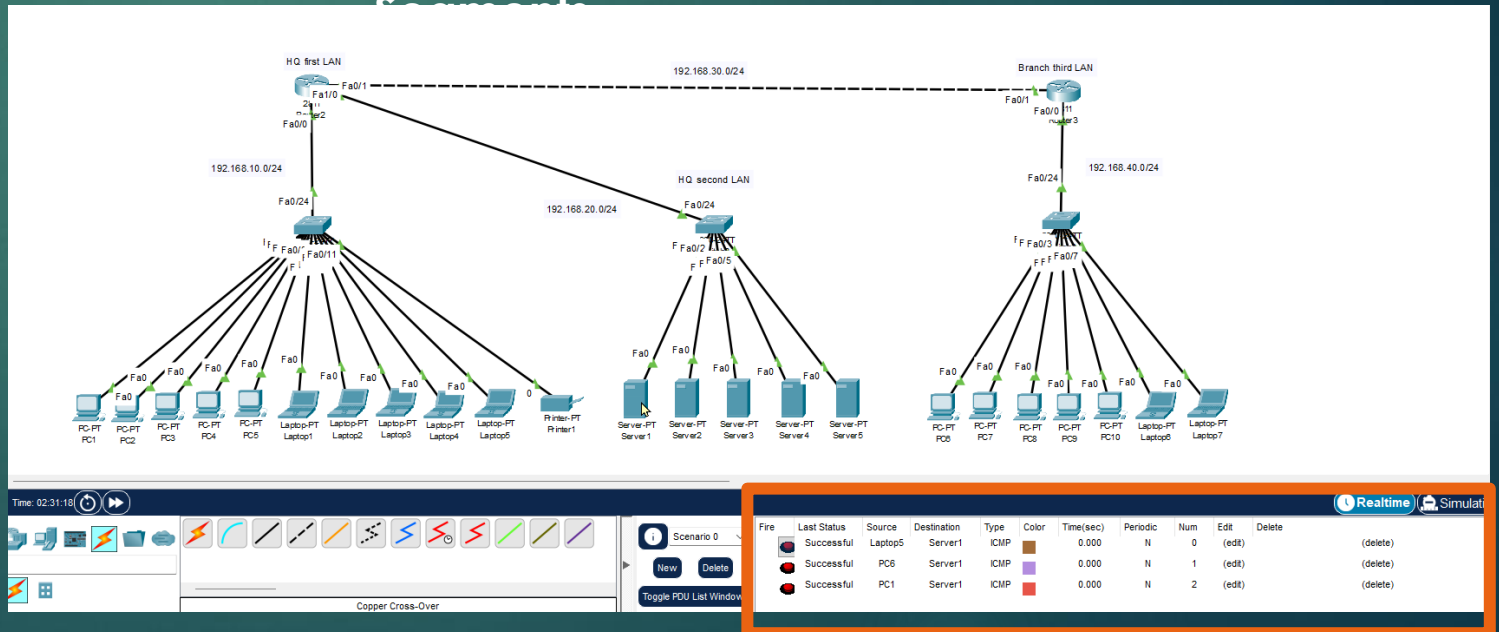
Project 1: Designing the Network Solution

Part 2: Network Expansion

Network Routing Proposal

I chose dynamic routing - **EIGRP** as preferred routing protocol because of its **fast convergence**, in case there are any changes of network topology, EIGRP can quickly find the best path for the data to reach the destination. Its **scalability** also makes it easier for future expansion, allow the network to expand without any major reconfiguration.

Ping Result Between All LAN



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Part 2: Network Expansion

Server Segmentation Justification

Why Server Are On a Separate LAN?

Server located on separated LAN segment **improves security** by reducing the chances of being expose to cyber threats and also helps to **increase available bandwidth** for users and backend. Its also allow **future scalability** without disrupting user access.

Project 1: Designing the Network Solution

Part 2: Network Expansion

HQ Router Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 100
Router(config-router)#net
% Incomplete command.
Router(config-router)#network 192.168.10.0
Router(config-router)#network 192.168.20.0
Router(config-router)#network 192.168.30.0
Router(config-router)#
```

Branch Router Configuration

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#network 192.168.30.0
^
% Invalid input detected at '^' marker.

Router(config)#router eigrp 100
Router(config-router)#network 192.168.30.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.30.1 (FastEthernet0/1) is up: new
adjacency

Router(config-router)#network 192.168.40.0
Router(config-router)#exit
Router(config)#exit
Router#
$SYS-5-CONFIG_I: Configured from console by console
```

“Up” Status of HQ LAN

```
Router>enable
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.10.254	YES	manual	up	up
FastEthernet0/1	192.168.20.254	YES	manual	up	up
FastEthernet1/0	192.168.30.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

“Up” Status of Branch LAN

```
Router>enable
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.40.254	YES	manual	up	up
FastEthernet0/1	192.168.30.2	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

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Project 1: Designing the Network Solution

Part 2: Network Expansion

Personal Reflection

Troubleshooting Journey and Experiences on Each Artifact Produced

During the process of setting up network routing, I encountered issue like **unable to ping between PCs and server**. After some troubleshooting, I realised that **wrong IP addresses** had been assigned to certain PCs, I reconfigured the IP setting to match the correct gateway. I also **facing issue on WAN connection** between HQ router and Branch router, the findings was I assigned the IP address to the **wrong port**. From this learning journey, I realise no matter how attentive or focus you are on work, always double check or maybe triple check before everything go live.



Thank You